

43. CANFD ECC (CNECC)

43.1 Overview

MBRAM have ECC function of 2-bit ECC error detection and 1-bit ECC error detection and correction^{*1}. The ECC module adds 7 bits ECC data to 32 bits RAM data.

Note 1. The ECC module cannot detect 3 or more bits error. In this case, the ECC module detects 1-bit or 2-bit error, does not detect errors, or corrects the erroneous bit to erroneous data by setting. When all RAM data are fixed to 0 or 1, it is detected as 2-bit ECC error.

43.2 Register Descriptions

43.2.1 EC710CTL : ECC Control Register

Base address: ECCMBn = 0x4036_F200 + 0x0100 × n (n = 0, 1)
ECCMBn_NS = 0x5036_F200 + 0x0100 × n (n = 0, 1)

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ECDE DF0	ECSE DF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EMCA[1:0]	—	—	—	ECOV FF	ECER 2C	ECER 1C	—	—	ECER VF	EC1E CP	EC2E DIC	EC1E DIC	ECER 2F	ECER 1F	ECER F
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
0	ECEMF	ECC Error Message Flag 0: There is no bit error in present RAM output data 1: There is bit error in present RAM output data	R
1	ECER1F	ECC Error Detection and Correction Flag 0: After clearing this bit, 1-bit error correction has not occurred 1: 1-bit error has occurred	R
2	ECER2F	2-bit ECC Error Detection Flag 0: After clearing this bit, 2-bit error has not occurred 1: 2-bit error has occurred	R
3	EC1EDIC	ECC 1-bit Error Detection Interrupt Control 0: Disable 1-bit error detection interrupt request 1: Enable 1-bit error detection interrupt request	R/W
4	EC2EDIC	ECC 2-bit Error Detection Interrupt Control 0: Disable 2-bit error detection interrupt request 1: Enable 2-bit error detection interrupt request	R/W
5	EC1ECP	ECC 1-bit Error Correction Permission 0: At 1-bit error detection, the error correction is executed 1: At 1-bit error detection, the error correction is not executed	R/W
6	ECERVF	ECC Error Judgment Enable Flag 0: Error judgment disable 1: Error judgment enable	R/W
8:7	—	These bits are read as 0. The write value should be 0.	R/W
9	ECER1C	Accumulating ECC Error Detection and Correction Flag Clear 0: No effect 1: Clear accumulating ECC error detection and correction flag	R/W
10	ECER2C	2-bit ECC Error Detection Flag Clear 0: No effect 1: Clear 2-bit ECC error detection flag	R/W

Bit	Symbol	Function	R/W
11	ECOVFF	ECC Overflow Detection Flag 0: No effect 1: ECC overflow detection flag	R
13:12	—	These bits are read as 0. The write value should be 0.	R/W
15:14	EMCA[1:0]	Access Control to ECC Mode Select bit These bits enable or disable write access to ECERVF bit.	R/W
16	ECSEDF0	ECC Single bit Error Address Detection Flag 0: There is no bit error in EC710EAD0 after reset or clearing ECER1F bit 1: Address captured in EC710EAD0 shows that 1-bit error occurred and captured	R
17	ECDEDF0	ECC Dual Bit Error Address Detection Flag 0: There is no bit error in EC710EAD0 after reset or clearing ECER2F bit 1: Address captured in EC710EAD0 shows that 2-bit error occurred and captured	R
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

ECEMF bit (ECC Error Message Flag)

The ECEMF bit shows that there is error in present read data bus. This bit is updated by every RAM output data.

When RAM output data is undefined and the ECERVF bit is set to 1, the value of this bit is undefined.

[Setting condition]

There is bit error in present RAM output data under the condition that error judgment is enabled.

[Clearing condition]

- Under the condition that there is no 1-bit error in input data to decode circuit
- When ECC error judgment is disabled (ECERVF = 0).

ECER1F bit (ECC Error Detection and Correction Flag)

The ECER1F bit shows that the bit errors are detected in the one part of RAM read data [38:0] at RAM read access when the error judgment is enabled.

When the 1-bit error interrupt output is enabled, error interrupt is generated by setting this flag.

This bit is read-only, so writing 1 or 0 has no effect.

At clearing, write 1 to the ECER1C bit.

When 1-bit error is detected again under the condition that this bit is set, the interrupt is not generated.

[Setting condition]

When the error judgment is enabled and there is 1-bit error to RAM output data (when not setting ECER1C = 1).

[Clearing condition]

- Writing ECER1C = 1
- When ECC error judgment is disabled (ECERVF = 0).

ECER2F bit (2-bit ECC Error Detection Flag)

The ECER2F bit shows that the bit errors are detected in the two parts of RAM read data [38:0] at RAM read access when the error judgment is enabled.

When the 2-bit error interrupt output is enabled, error interrupt is generated by setting this flag.

This bit is read-only, so writing 1 or 0 has no effect.

At clearing, write 1 to the ECER2C bit.

When 2-bit error is detected again under the condition that this bit is set, the interrupt is not generated.

[Setting condition]

When the error judgment is enabled and there is 2-bit error to RAM output data (when not setting ECER2C = 1).

[Clearing condition]

- Writing ECER2C = 1
- When ECC error judgment is disabled (ECERVF = 0).

EC1EDIC bit (ECC 1-bit Error Detection Interrupt Control)

The EC1EDIC controls the interrupt output at detecting 1-bit error. By setting 1 to this bit, the 1-bit error interrupt is outputted when 1-bit error detected.

EC2EDIC bit (ECC 2-bit Error Detection Interrupt Control)

The EC2EDIC controls the interrupt output at detecting 2-bit error. By setting 1 to this bit, the 2-bit error interrupt is outputted when 2-bit error detected.

EC1ECP bit (ECC 1-bit Error Correction Permission)

The EC1ECP sets enable or disable to correct the 1-bit error when ECC error detection and correction is valid. By setting 1 to this bit, the non-corrected data is outputted if 1-bit error is detected.

ECERVF bit (ECC Error Judgment Enable Flag)

Setting the ECERVF bit to 1 enables the judgment of error. The correction of output data and the interrupt output depend on setting of the EC1ECP bit, EC2EDIC bit, and EC1EDIC bit.

The write access to this bit is valid when the write value of the EMCA[1:0] is 01b. So only the 16 bits or 32 bits operation command is valid in the case of the write access to this bit.

ECER1C bit (Accumulating ECC Error Detection and Correction Flag Clear)

The ECER1C bit clears the status flag of the ECER1F bit.

The read value is always 0. By writing 0, the internal condition is not changed. When the competition between writing 1 to this bit and setting the ECER1F bit, the former has priority.

The ECER1F bit is cleared by writing 1 to this bit while the ECER1F bit is set. Additionally, the Overflow Detection flag (ECOVFF), ECC Dual Bit Error flag (ECDEDF0) and ECC Single Bit Error flag (ECSEDF0) are also cleared.

ECER2C bit (2-bit ECC Error Detection Flag Clear)

The ECER2C bit clears the status flag of the ECER2F bit.

The read value is always 0. By writing 0, the internal condition is not changed. When the competition between writing 1 to this bit and setting the ECER2F bit, the former has priority.

The ECER2F bit is cleared by writing 1 to this bit while the ECER2F bit is set. Additionally, the Overflow Detection flag (ECOVFF), ECC Dual Bit Error flag (ECDEDF0), and ECC Single Bit Error flag (ECSEDF0) are also cleared.

ECOVFF bit (ECC Overflow Detection Flag)

The ECOVFF bit is set and the overflow interruption is outputted by detecting the new error address under the condition that error address is already captured in the EC710EAD0 register. The overflow interrupt is outputted again when this bit is set and new error is detected.

This bit is read-only, so writing 1 or 0 has no effect.

To clear this bit, write 1 to the ECER2C bit and the ECER1C bit.

[Setting condition]

When new error address is captured under the condition that error address is already captured in the EC710EAD0 register (when not setting ECER2C = 1 or ECER1C = 1).

[Clearing condition]

- Writing ECER2C = 1 and ECER1C = 1
- When ECC error judgment is disabled (ECERVF = 0).

EMCA[1:0] bit (Access Control to ECC Mode Select bit)

The EMCA[1:0] bits are the write trigger reserved bits to the ECERVF bit. The read value is always 0. When the value of these bits is 01b, it is possible to have write access to the ECERVF bit. If these bits are not 01b, write access to the ECERVF bit is ignored and the value is not written.

ECSEDF0 bit (ECC Single bit Error Address Detection Flag)

The ECSEDF0 bit shows that the error is captured in the error address register when error detection is valid. This bit is set by 1-bit error detection.

When 1-bit error is detected after the 2-bit error address is already captured in the EC710EAD0 register, this bit is not updated and the EC710EAD0 register is not updated.

This bit is read-only, so writing 1 or 0 has no effect. To clear this bit, write 1 to the ECER2C bit and the ECER1C bit.

[Setting condition]

When there is 1-bit error to RAM output data and error address is captured in EC710EAD0 under the condition that the error judgment is permitted (when not setting ECER1C = 1).

[Clearing condition]

- Writing ECER2C = 1 and ECER1C = 1
- When ECC error judgment is disabled (ECERVF = 0).

ECDEDF0 bit (ECC Dual Bit Error Address Detection Flag)

The ECDEDF0 bit shows that the error is captured in the error address register when error detection is valid. This bit is set by 2-bit error detection.

When 2-bit error is detected after the 1-bit error address is already captured in the EC710EAD0 register, this bit is not updated but the EC710EAD0 register is updated.

This bit is read-only, so writing 1 or 0 has no effect. To clear this bit, write 1 to the ECER2C bit and the ECER1C bit.

[Setting condition]

When there is 2-bit error to RAM output data and error address is captured in EC710EAD0 under the condition that the error judgment is permitted (when not setting ECER2C = 1).

[Clearing condition]

- Writing ECER2C = 1 and ECER1C = 1
- When ECC error judgment is disabled (ECERVF = 0).

43.2.2 EC710TMC : ECC Test Mode Control Register

Base address: ECCMBn = 0x4036_F200 + 0x0100 × n (n = 0, 1)
ECCMBn_NS = 0x5036_F200 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ETMA[1:0]	—	—	—	—	—	—	ECTM CE	—	—	—	—	—	—	ECDC S	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	ECDCS	ECC Decode Input Select 0: Input lower 32 bits of RAM output data to data area of decode circuit 1: Input ECEDB31-0 in EC710TED register to data area of decode circuit	R/W
6:2	—	These bits are read as 0. The write value should be 0.	R/W
7	ECTMCE	ECC Test Mode Control Enable 0: The access to test mode register and bit is disabled 1: The access to test mode register and bit is enabled	R/W
13:8	—	These bits are read as 0. The write value should be 0.	R/W
15:14	ETMA[1:0]	ECC Test Mode Bit Access Control These bits enable or disable write access to ECTMCE bit.	R/W

Note: S-TYPE-3, P-TYPE-3

ECDCS bit (ECC Decode Input Select)

The ECDCS bit selects either the lower 32 bits data value from RAM or value from the internal test register (EDEDDB[31:0] in EC710TED) as input signal to decoder.

The write access to this bit is valid under the condition of ECTMCE = 1 (it is possible to set them at the same time.)

This bit is cleared by setting ECTMCE = 0.

ECTMCE bit (ECC Test Mode Control Enable)

The ECTMCE bit selects the access enable or disable to test register and test control bit.

The write access to this bit is valid under the condition that the value of the ETMA[1:0] bits is 10b.

ETMA[1:0] bits (ECC Test Mode Bit Access Control)

The ETMA[1:0] bits are the write trigger reserved bits to the ECTMCE bit. The read value is always 0. When the value of these bits is 10b, it is possible to have write access to the ECTMCE bit. If these bits are not 10b, the write access to the ECTMCE bit is ignored and the value is not written.

43.2.3 EC710TED : ECC Test Substitute Data Register

Base address: ECCMBn = 0x4036_F200 + 0x0100 × n (n = 0, 1)
ECCMBn_NS = 0x5036_F200 + 0x0100 × n (n = 0, 1)

Offset address: 0x0C

Bit position: 31

0

Bit field:

ECEDB[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	ECEDB[31:0]	ECC Test Substitute Data Substitute data in ECC test mode.	R/W

Note: S-TYPE-3, P-TYPE-3

This register is for the 32 bits data for ECC decode. It is possible to read and write using 32 bits operation command when ECTMCE = 1. When ECTMCE = 0, all bits are always 0.

ECEDB[31:0] bits (ECC Test Substitute Data)

When ECDCS in EC710TMC register is 1, the value of this register is bits [31:0] of the input data to the decode circuit.

43.2.4 EC710EAD0 : ECC Error Address Register

Base address: ECCMBn = 0x4036_F200 + 0x0100 × n (n = 0, 1)
ECCMBn_NS = 0x5036_F200 + 0x0100 × n (n = 0, 1)

Offset address: 0x10

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	ECEAD[9:0]									
---	---	---	---	---	---	------------	--	--	--	--	--	--	--	--	--

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
9:0	ECEAD[9:0]	ECC Error Address	R

Bit	Symbol	Function	R/W
31:10	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

This is a read-only register to hold the ECC error address.

ECEAD[9:0] bits (ECC Error Address)

When ECC error is detected for permitting ECC error judgment, RAM address is captured by the detected signal as a trigger and is hold as the error occurring address. The error address is not captured when the error occurred again to the one held by the same factor.

If 2-bit error occurred under the condition that 1-bit error address is already captured, the 2-bit error address is over-written but the ECDEDF0 bit is not set to 1.

If 1-bit error occurred under the condition that 2-bit error address is already captured, the 1-bit error address is not overwritten and the ECSEDF0 bit is not set to 1.

This register is reset by clearing the status flag (the ECER2F bit or the ECER1F bit).

43.3 Operation

43.3.1 ECC Function Setting

Figure 43.1 shows a procedure for ECC function setting.

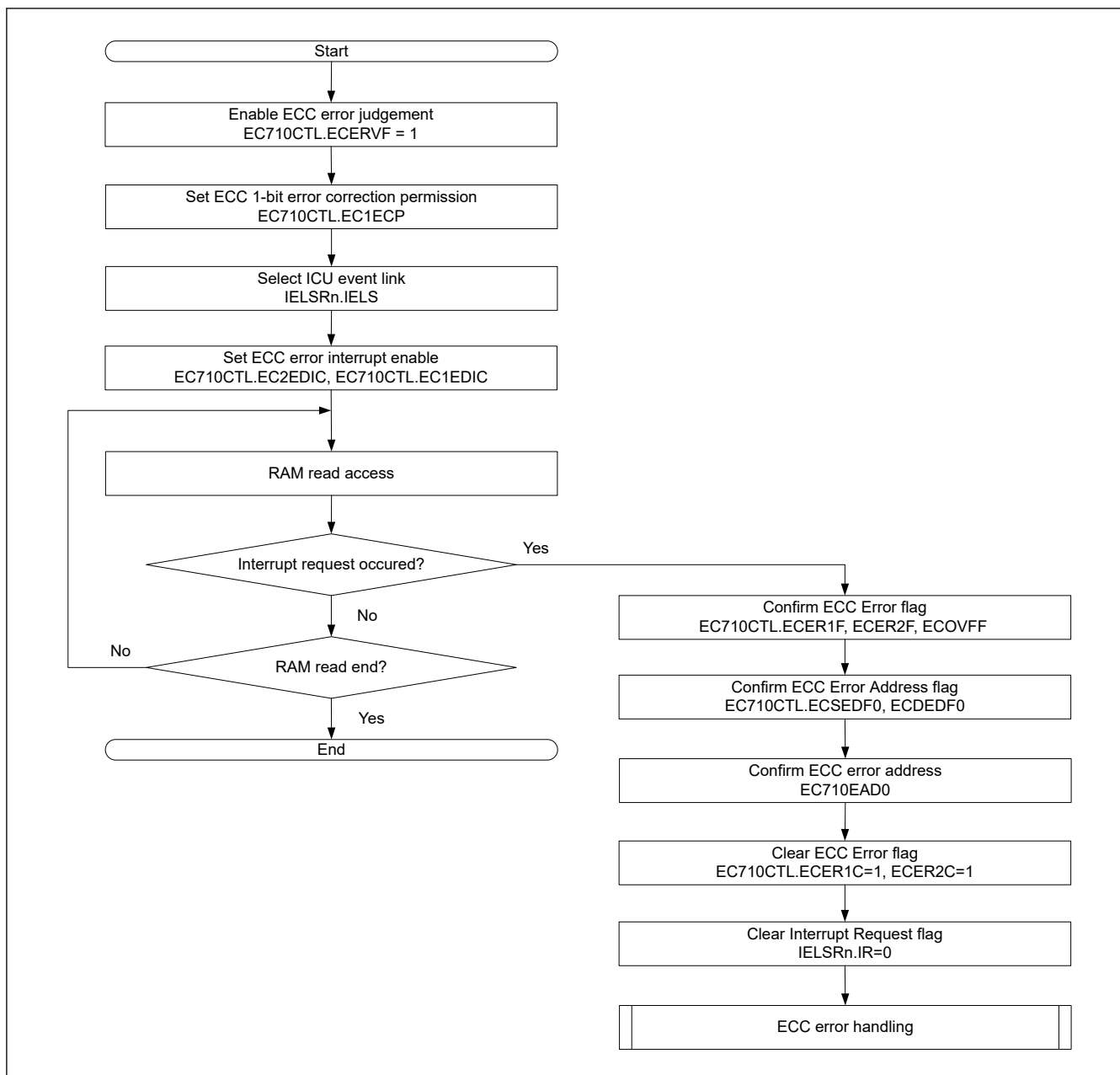


Figure 43.1 Setting procedure for ECC function

43.3.2 ECC Decoder Testing

ECC interrupts can be intentionally generated by ECC test mode. [Figure 43.2](#) shows a procedure for ECC decoder testing.

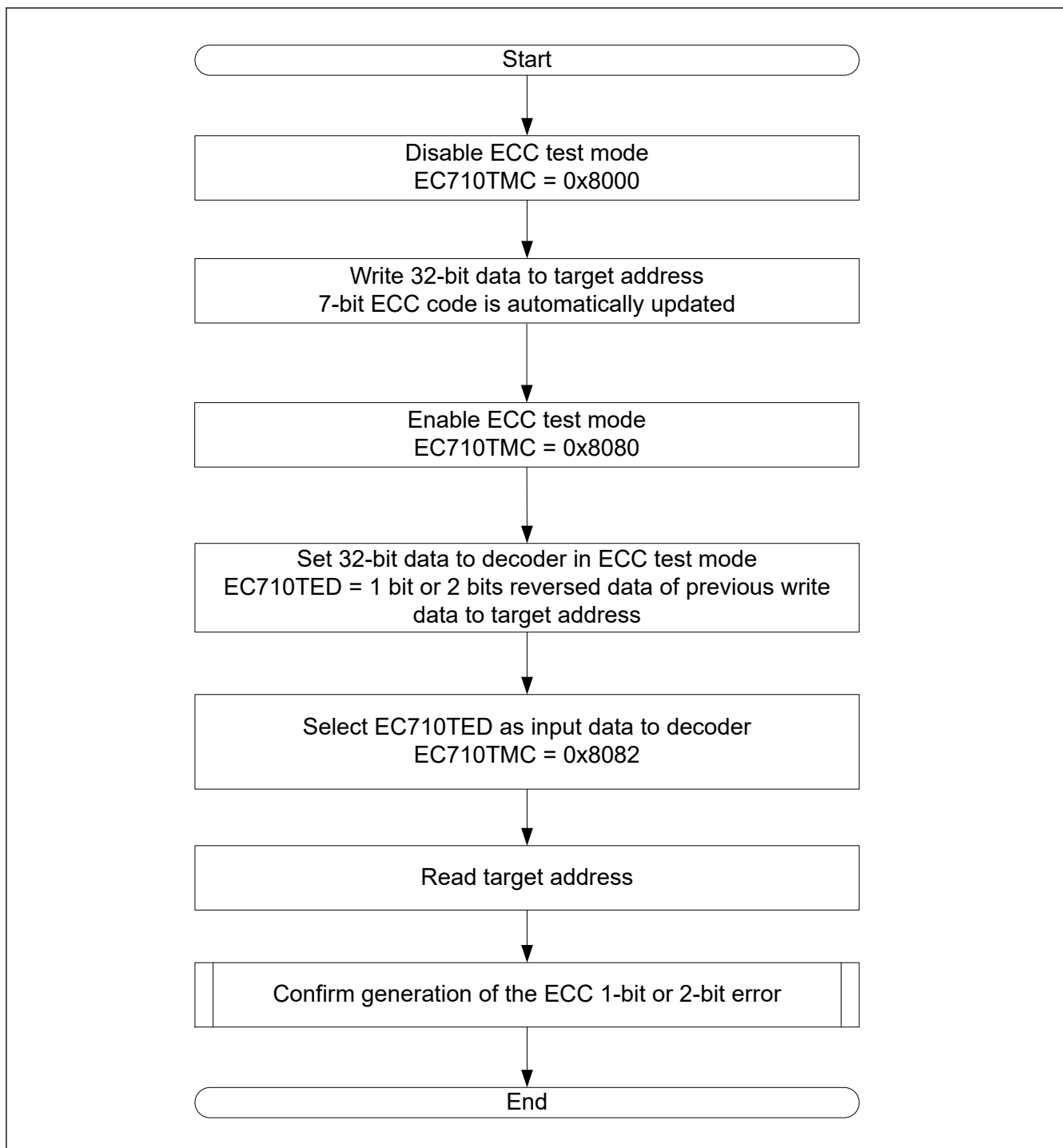


Figure 43.2 Testing procedure for ECC decoder

43.4 Interrupts

The ECC module issues three interrupt requests:

- CANn_MRAM_ERI (n = 0, 1)

Interrupt sources of each interrupt request include:

- 1-bit ECC error
- 2-bit ECC error
- ECC error overflow.

43.5 Usage Notes

43.5.1 Enter Software Standby Mode

The following steps are required, before the MCU enters Software Standby mode.

1. To clear the ECC error flags and ECC error address, write 1 to the ECER2C bit and the ECER1C bit.
2. To disable ECC error judgment, write 0 to the ECERVF bit.

43.5.2 Return from Software Standby Mode

The following steps are required, after the MCU returns from Software Standby mode.

1. Make initial settings.
2. To enable ECC error judgment, write 1 to the ECERVF bit.